

Blockchain Technology

Fundamentals and Core Concepts

Suresh Paul

Algorithm Basics and BW

What We'll Cover Today

1.  What is blockchain? Core concepts
2.  Brief history of blockchain
3.  Key features and characteristics
4.  Benefits and drawbacks
5.  Hands-on: Exploring a blockchain

What is Blockchain?

Simple Definition:

- ▶  A **digital ledger** (record book) shared across many computers
- ▶  Records are **permanent** and cannot be changed
- ▶  No single person or company controls it
- ▶  Transactions are grouped into “blocks” and linked together

Think of it like:

- ▶ A  Google Doc that everyone can read
- ▶ But once you write something, you can't erase it
- ▶ And no single person owns the document

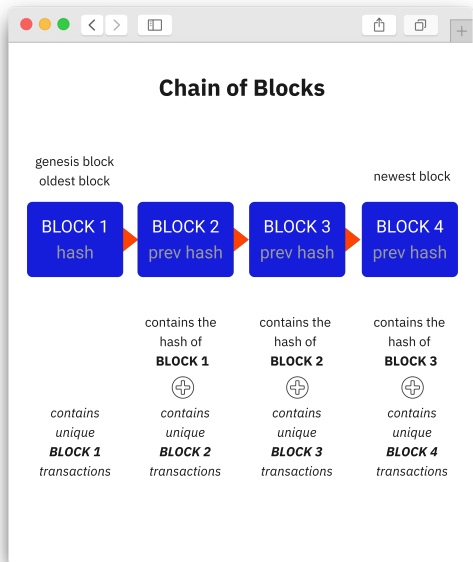
Blockchain Visualized

How blocks connect:

- ▶  Each block = container of transactions
- ▶  Blocks link to previous block
- ▶  Forms a chain (blockchain!)
- ▶  Changing old blocks = impossible

Key concept:

- ▶ Like a tower of blocks
- ▶ Can't remove bottom blocks without toppling



Traditional Database vs Blockchain

Feature	Traditional Database	Blockchain
Control	One company/person	Distributed (many people)
Changes	Can edit/delete	Cannot change history
Trust	Trust the company	Trust the system/math
Access	Permission needed	Often public
Speed	Very fast	Slower (presently)

When to use each:

- ▶  Traditional: Company internal data, customer records
- ▶  Blockchain: Public records, financial transactions, voting

A Brief History of Blockchain

1. **2008:**  Satoshi Nakamoto publishes Bitcoin whitepaper
2. **2009:** First Bitcoin transaction (Genesis Block)
3. **2015:**  Ethereum launches - adds “smart contracts”
4. **2017:** ICO boom - thousands of new cryptocurrencies
5. **2018-2024:**  DeFi, NFTs, enterprise adoption grows
6. **2026:**  Strategic Bitcoin Reserve ???

Key milestone:

- ▶  Bitcoin solved the “double-spending” problem
- ▶ First time digital money worked without a bank

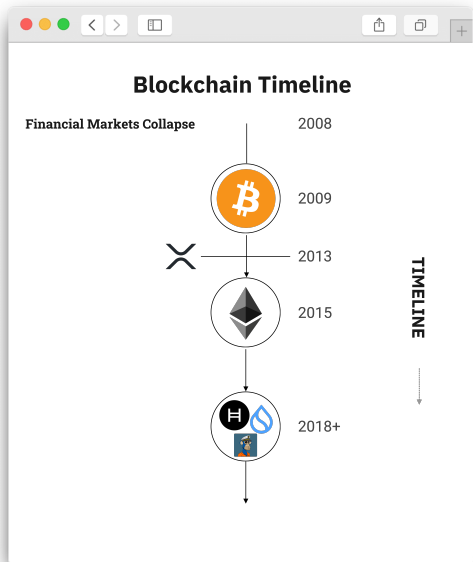
Timeline Visualization

Evolution of blockchain:

- ▶ **Gen 1:** ₿ Bitcoin
 - Digital gold / money
- ▶ **Gen 2:** ⬡ Ethereum
 - Programmable blockchain
- ▶ **Gen 3:** Modern chains
 - Faster, cheaper

From:

- ▶ Peer-to-peer payments
- ▶ → Smart contracts
- ▶ → Full applications on-chain



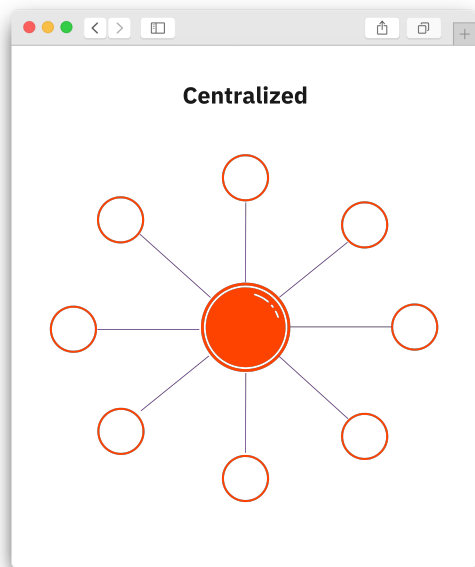
Core Characteristics of Blockchain

1.  **Immutability:** Can't change past records
2.  **Decentralization:** No single point of control
3.  **Transparency:** Everyone can see transactions
4.  **Security:** Cryptography protects data
5.  **Consensus:** Network agrees on truth

Real-world analogy:

- ▶ Like a town hall meeting where everyone votes
- ▶ Everything said is recorded permanently
- ▶ No mayor makes all decisions alone

Decentralization Explained



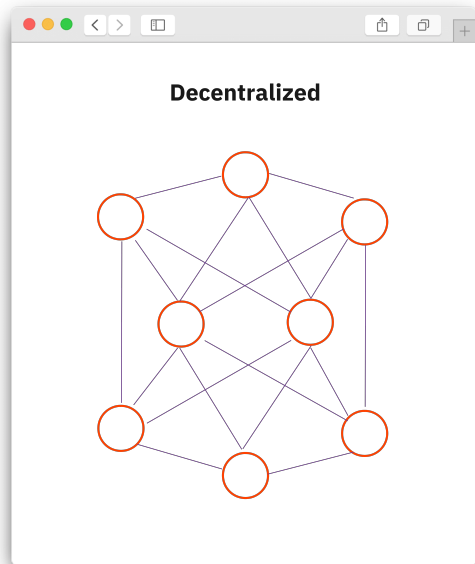
Centralized (Traditional):

- ▶ 🏢 One company controls
- ▶ Single point of failure
- ▶ Faster decisions

Decentralized (Blockchain):

- ▶ 👥 Many participants
- ▶ No single point of failure
- ▶ Democratic but slower

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How Transactions Work

Step-by-step process:

1. 📌 Alice wants to send money to Bob
2. 'A' Transaction broadcast to network
3. ✅ Network validates transaction (is it legit?)
4. ■ Transaction added to a new block
5. ⚙️ Block added to the blockchain
6. ✔️ Bob receives the money

Time: Usually 10 seconds to 10 minutes depending on blockchain

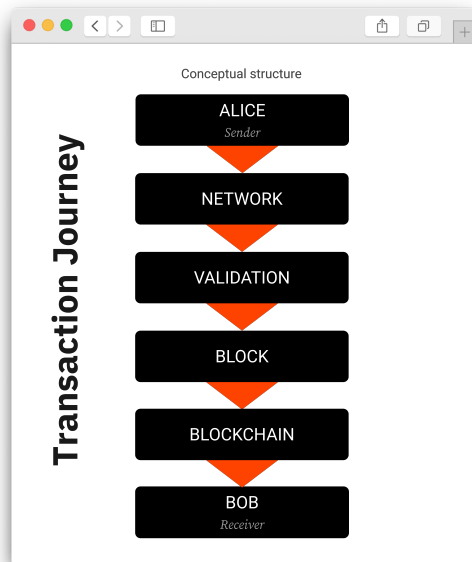
Transaction Flow Diagram

What happens:

- ▶  Sender initiates
- ▶  Network verifies
- ▶  Miners/validators confirm
- ▶  Added to blockchain
- ▶  Receiver gets funds

Key point:

- ▶ No bank in the middle!
- ▶ Network does the verification



Benefits of On-Chain Applications

1.  **Global access:** Anyone with internet can participate
2.  **Censorship resistant:** No one can shut it down
3.  **Transparency:** All activity is public and auditable
4.  **Automation:** Smart contracts execute automatically
5.  **Lower fees:** No middlemen taking cuts (sometimes)

Example use cases:

- ▶ International money transfers
- ▶ Supply chain tracking
- ▶ Digital identity systems

Drawbacks of On-Chain Applications

1. 🕒 **Speed:** Slower than traditional databases
2. 💰 **Cost:** Can be expensive during high usage
3. 🎓 **Complexity:** Harder for average users
4. ⚠️ **Irreversible:** Mistakes can't be undone
5. ⚡ **Energy:** Some blockchains use lots of electricity

Trade-offs:

- ▶ Decentralization → Slower speed
- ▶ Security → Higher cost
- ▶ Transparency → Less privacy

Blockchain Trilemma

The fundamental challenge:

- ▶  **Security:** Protection from attacks
- ▶  **Decentralization:** Many participants
- ▶ **X¹ Scalability:** Handle many transactions

You can typically only optimize 2 out of 3!

Examples:

- ▶  Bitcoin: High security + decentralization = Low scalability
- ▶ Some new chains: High speed + scalability = Less decentralized

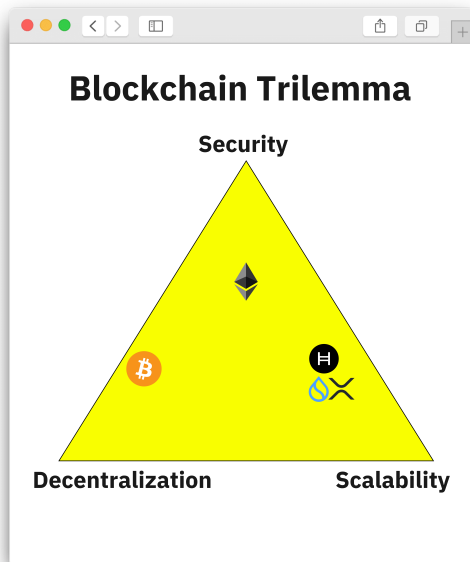
Trilemma Visualization

Pick two:

- ▶ Can't have all three perfectly
- ▶ Different blockchains make different trade-offs
- ▶ No “perfect” solution yet

Current solutions:

- ▶ Layer 2 networks
- ▶ Sharding
- ▶ New consensus mechanisms



Types of Blockchains

1.  **Public:** Anyone can join (🔶 Bitcoin, ⬥ Ethereum)
2.  **Private:** Permission required (Company internal)
3.  **Consortium:** Group of organizations control
4.  **Hybrid:** Mix of public and private

When to use each:

- ▶ Public: Cryptocurrencies, open finance
- ▶ Private: Supply chain, internal auditing
- ▶ Consortium: Bank settlements, industry cooperation

Smart Contracts: Code on Blockchain

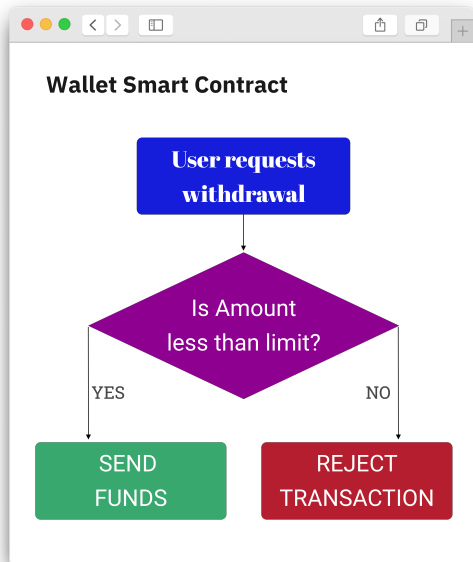
What are smart contracts?

- ▶  Programs that run on blockchain
- ▶  Execute automatically when conditions met
- ▶  Like a digital vending machine
- ▶ No lawyers or middlemen needed

Example:

- ▶ “IF payment received, THEN transfer ownership”
- ▶ “IF date = January 1, THEN pay rent automatically”
- ▶ Code enforces the agreement

Smart Contract Example



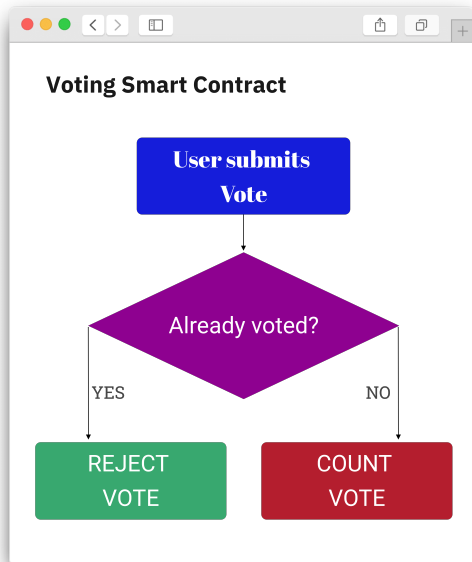
Traditional contract:

- ▶ Written on paper
- ▶ Lawyers enforce
- ▶ Can be disputed
- ▶ Takes time

Smart contract:

- ▶ Written in code
- ▶ Code enforces
- ▶ Automatic execution
- ▶ Instant (minutes)

Smart Contract Example



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Hands-On Activity: Blockchain Explorer

Let's explore a real blockchain!

We'll use: `https://etherscan.io`

1.  Open browser and go to Etherscan
2.  Look at the latest block
3.  See recent transactions
4.  Check transaction amounts
5.  Explore wallet addresses

What you'll see:

- ▶ Real transactions happening right now
- ▶ All data is public and transparent

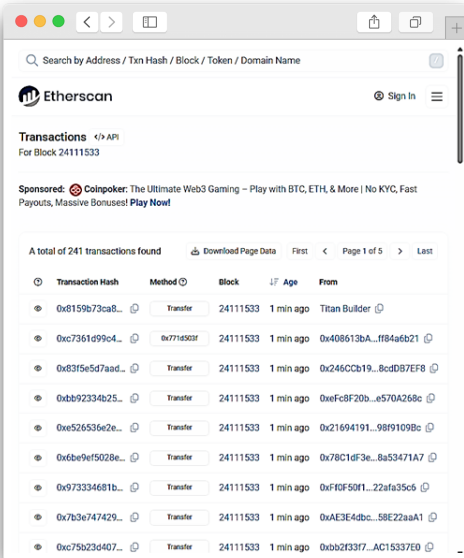
Exploring Etherscan - Step by Step

What to look for:

1.  Block number
2.  Timestamp
3.  From/To addresses
4.  Amount transferred
5.  Gas fee (transaction cost)

Try this:

- ▶ Click on a transaction
- ▶ See all the details
- ▶ Notice: everything is public!



The screenshot shows the Etherscan website interface. At the top, there's a search bar with the text "Search by Address / Txn Hash / Block / Token / Domain Name". Below the search bar is the Etherscan logo and a "Sign In" button. The main heading is "Transactions" with a sub-heading "For Block 24111533". There's a sponsored advertisement for "Coinpoker" below the heading. The main content area displays a table of transactions. The table has columns for "Transaction Hash", "Method", "Block", "Age", and "From". There are 241 transactions found in total. The first transaction is a "Transfer" with hash "0x8159b73ca8..." from "Titan Builder". The second transaction is a "Transfer" with hash "0xc771d503f..." from "0x408613bA...ff84a6b21". The third transaction is a "Transfer" with hash "0x83f5e5d7aad..." from "0x246CCb19...8cdB87EF8". The fourth transaction is a "Transfer" with hash "0xbb92334b25..." from "0xeFc8F20b...e570A268c". The fifth transaction is a "Transfer" with hash "0xe526536e2e..." from "0x21694191...98f9109Bc". The sixth transaction is a "Transfer" with hash "0x6be9ef5028e..." from "0x78C1dF3e...8a53471A7". The seventh transaction is a "Transfer" with hash "0x973334681b..." from "0xF0F50f1...22afa35c6". The eighth transaction is a "Transfer" with hash "0xb3e747429..." from "0xAE3E4dbc...58E22aaA1". The ninth transaction is a "Transfer" with hash "0xc75b23d407..." from "0xbb2f33f7...AC15337E0".

Transaction Hash	Method	Block	Age	From
0x8159b73ca8...	Transfer	24111533	1 min ago	Titan Builder
0xc771d503f...	Transfer	24111533	1 min ago	0x408613bA...ff84a6b21
0x83f5e5d7aad...	Transfer	24111533	1 min ago	0x246CCb19...8cdB87EF8
0xbb92334b25...	Transfer	24111533	1 min ago	0xeFc8F20b...e570A268c
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0x973334681b...	Transfer	24111533	1 min ago	0xF0F50f1...22afa35c6
0xb3e747429...	Transfer	24111533	1 min ago	0xAE3E4dbc...58E22aaA1
0xc75b23d407...	Transfer	24111533	1 min ago	0xbb2f33f7...AC15337E0

Understanding a Block

Key information in every block:

1. **# Block number:** Sequential identifier
2. **🕒 Timestamp:** When block was created
3. **↔ Transactions:** Number of transactions included
4. **👤 Miner/Validator:** Who created this block
5. **📄 Gas used:** Total fees paid

Fun fact:

- ▶ New **💎** Ethereum block every 12 seconds
- ▶ **₿** Bitcoin block every 10 minutes

Key Takeaways

1.  Blockchain = Shared, permanent digital ledger
2.  Once written, cannot be changed (immutable)
3.  No single entity controls it (decentralized)
4.  Trade-offs between speed, cost, and decentralization
5.  Smart contracts = programmable agreements

Remember:

- ▶ Blockchain is a tool, not a solution for everything
- ▶ Best for situations requiring trust and transparency

Questions and Discussion